

```
#include <stdio.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/uart.h"
```

```
#define BUF_SIZE      (1024)
#define UART_RX_PIN   (3)
#define UART_TX_PIN   (1)
```

```
void app_main(void)
{
    xTaskCreate(echo_task, "uart_echo_task", CONFIG_EXAMPLE_TASK_STACK_SIZE, NULL, 10, NULL);
}
```

```
static void echo_task(void *arg)
{
    uart_config_t uart_config = {
        .baud_rate = 115200,
        .data_bits = UART_DATA_8_BITS,
        .parity = UART_PARITY_DISABLE,
        .stop_bits = UART_STOP_BITS_1,
        .flow_ctrl = UART_HW_FLOWCTRL_DISABLE,
        .source_clk = UART_SCLK_DEFAULT,
    };
    // Configure UART parameters
    ESP_ERROR_CHECK(uart_param_config(UART_NUM_0, &uart_config));

    // Set UART pins
    ESP_ERROR_CHECK(uart_set_pin(UART_NUM_0, UART_TX_PIN, UART_RX_PIN, UART_PIN_NO_CHANGE, UART_PIN_NO_CHANGE));

    // Install UART driver
    ESP_ERROR_CHECK(uart_driver_install(UART_NUM_0, BUF_SIZE * 2, \
                                       BUF_SIZE * 2, 0, NULL, 0));

    // Configure a temporary buffer for the incoming data
    uint8_t *data = (uint8_t *) malloc(BUF_SIZE);

    while (1) {
        // Read data from the UART
        int len = uart_read_bytes(UART_NUM_0, data, (BUF_SIZE - 1), 20 / portTICK_PERIOD_MS);
        // Write data back to the UART
        uart_write_bytes(UART_NUM_0, (const char *) data, len);
    }
}
```